



Oscar Breiner

M.Sc. Informatics Student (TUM), AI & Perception Researcher

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Research Experience

4D Radar Perception in Autonomous Racing

- A2RL Autonomous Racing World Champion (2025)
- Built a real-time 4D radar perception and localization pipeline
- Developed radar-only odometry with a novel Doppler de-aliasing method for robust ego-motion at 250+ km/h
- IROS 2026 submission pending

3D Vision Reconstruction (Prof. Dai)

- Improved SfM-free 3D Gaussian Splatting for large-surface rendering
- Guided MCMC with blending visibility and error signals

Multimodal Model Merging (Prof. Cremers)

- Developed multimodal model merging via geometric alignment
- Improved efficiency via weight preconditioning and truncation

AI Team Lead – WARR Student Club at TUM

- Contributed to satellite mission launched in 2024
- Led team of 5, developing ML for space debris impact prediction

Additional Research Projects

- GNN Uncertainty & Vision Transformer Review (Prof. Günnemann)
- B.Sc.: DL for Space Debris Detection using Radar (Prof. Günnemann)
- Explored physics-informed learning (PINNs, hybrid methods)

Work Experience

BMW Group

Sep 2024 – Present

Machine Learning Engineer

- Improved performance of multimodal ML systems
- Built MLOps infrastructure and advised on ML modeling and evaluation

Radar Detection Startup (Actlabs)

Oct 2023 – Mar 2024

Bachelor's Thesis (TUM Collaboration)

- Developed DL methods for radar-based space debris detection
- Simulated I/Q radar signals based on radar cross-section models

Bosch Siemens Home Appliances (BSH)

Oct 2022 – Feb 2023

iOS Software Engineer

- Developed an iOS app from scratch using Swift and Google Firebase
- Implemented 3D scanning, mapping, and augmented reality features

Technical University of Munich (TUM)

Oct 2021 – Aug 2022

Software Engineering Teaching Assistant

- Graded 1,000+ exams; taught weekly reviews; mentored teams

Education – Technical University of Munich

TUM M.Sc. Informatics

Apr 2024 – Present

German scale grade: 1.4 (to date)

TUM B.Sc. Informatics

Oct 2020 – Mar 2024

Minor in Electrical Engineering, German scale grade: 2.0

Links

LinkedIn

linkedin.com/in/oscarbreiner

Portfolio & GitHub

oscarbreiner.github.io

Technical Skills

- AI in Vision, Robotics & Physics
- Modalities: Vision, Radar, Lidar, Text, Graphs, Time Series
- Robotics: ROS2, Perception
- Deep Learning & Simulation
- Python (+ PyTorch), Java, C++, Assembly, Swift, SQL
- Git, Docker, Linux, Matlab
- MLOps: AWS, Slurm, Wandb

Languages

English (Bilingual, C2)

German (Native)

French (B1), Japanese (Basics)

Certificates

Deutschlandstipendium

German National Scholarship, Top 1.2% nationwide

German Physical Society Award

CrossFit Coach License

Passions

- Competitive Rowing
- Olympic Weightlifting